DO NOW — RECALL

# Lesson 082 · Quicksort: Partition Around a Pivot

Name: _____ Class: _____ Date: _____

**1. What is merge sort's combine step?**

**2. What is a pivot in quicksort?**

**3. Is a partitioned array necessarily fully sorted?**

**4. Swap values at indexes0 and2 in [4,1,7]. Give the resulting array.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Merge two individually sorted runs.

### 2. Model answer

Selected value used to partition an array.

### 3. Model answer

No; subarrays still need sorting.

### 4. Model answer

[7,1,4].

### By the end, you can…

• Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

• The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

• Average and best-case time O(n log n), worst O($n^2$) with poor pivot choices; recursion consumes stack space depending on partition balance.

### Vocabulary — know these terms

quicksort — Recursive partition-based sorting.

pivot — Value partitioning subarray.

partition — Place smaller/larger values around pivot.

worst case — Maximum growth for adverse input.

NEW LEARNING

# Read, annotate and explain

## Observe

Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

## Apply

The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

## Evaluate

Average and best-case time O(n log n), worst O($n^2$) with poor pivot choices; recursion consumes stack space depending on partition balance.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

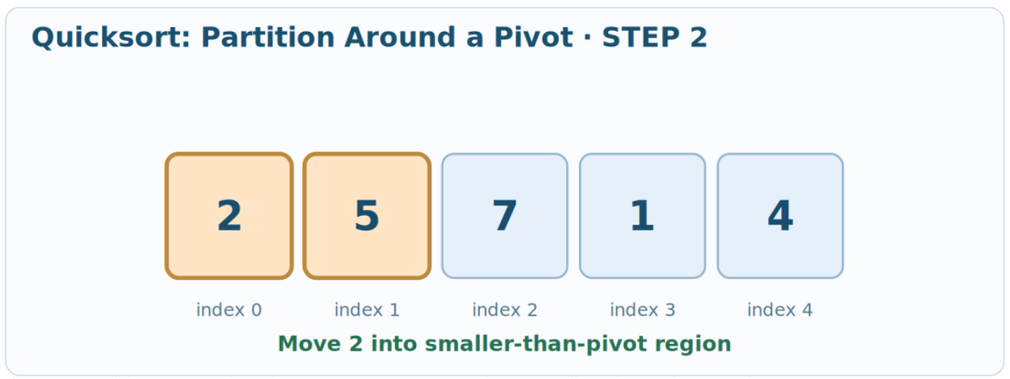
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Quicksort: Partition Around a Pivot · STEP 1; Choose last pivot4

**New Learning**




Quicksort: Partition Around a Pivot · STEP 2; Move 2 into smaller-than-pivot region



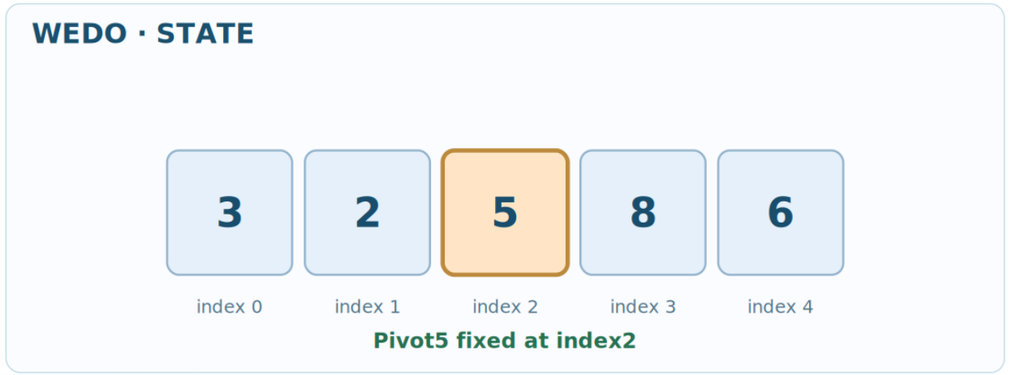


# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Partition [8,3,6,2,5] using last pivot5.**

Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].




WEDO · STATE; Pivot5 fixed at index2

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

Method and result
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

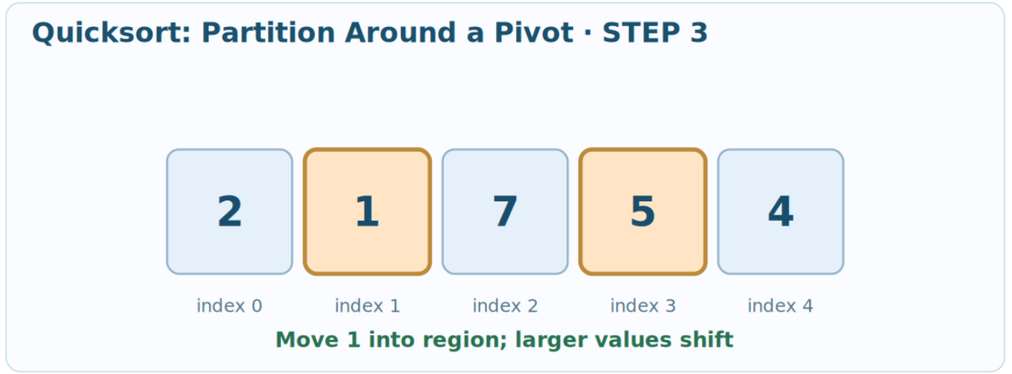




# New Learning · original lesson diagrams

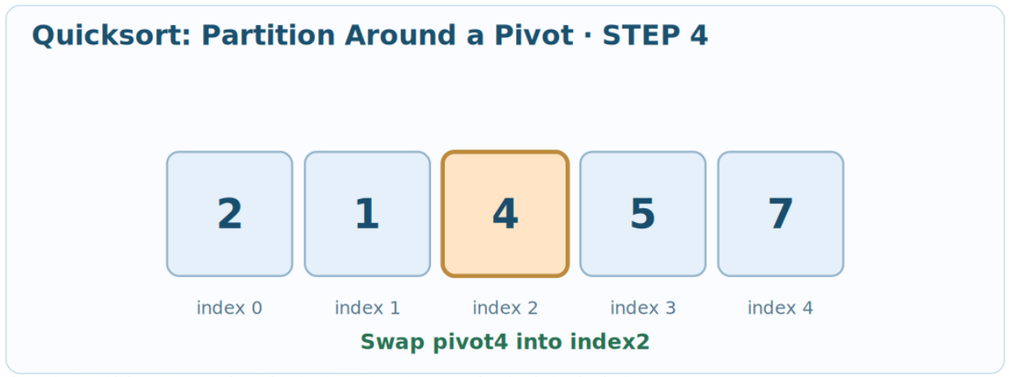
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Quicksort: Partition Around a Pivot · STEP 3; Move 1 into region; larger values shift

**New Learning**




Quicksort: Partition Around a Pivot · STEP 4; Swap pivot4 into index2

I DO — WORKED EXAMPLE

# Follow the worked method

### Partition [6,2,5,1,3] using last pivot3; give partition state.

### Method and result

Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Once a pivot reaches its correct final index, both partitions are also completely sorted.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

### GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Explain how pivot choice impacts quicksort worst-case time and recursive stack depth. [6]



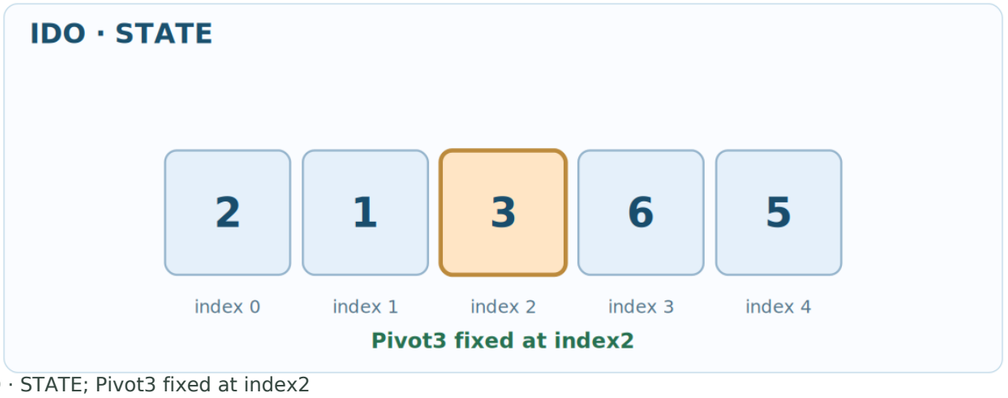


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Partition [6,2,5,1,3] using last pivot3; give partition state.**

Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].




IDO · STATE; Pivot3 fixed at index2

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Partition [8,3,6,2,5] using last pivot5.
Source code / input:
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Trace Lomuto pivot-last partition of [5,2,7,1,4] and give final pivot index. [6]

### Question 2

After quicksort partition gives [2,1,4,5,7] with pivot4 at index2, state required recursive subranges and eventual sorted result. [6]